Lesson 33: Arc Length of Polar Curve

Criteria	🌟 Exemplary (4)	🚀 Proficient (3)	⚙️ Developing (2)	⚠️ Emerging (1)
Setup & Formula	Correctly identifies r and $dr/d\theta$. Sets up the integral $\int \sqrt{r^2 + (r')^2}\, d\theta$ perfectly.	Correct formula used, but minor error in r or $dr/d\theta$. Setup is generally correct.	Formula recalled incorrectly, or major errors in identifying r or $dr/d\theta$.	Misunderstands the formula; attempts to use Cartesian arc length.
Limits of Integration	Identifies the exact interval $[\alpha, \beta]$ to trace the curve exactly once.	Identifies correct bounds, but may trace the curve twice (e.g., $0 \rightarrow 2\pi$ instead of $0 \rightarrow \pi$).	Misinterprets the graph; incorrect upper/lower limits.	No understanding of interval limits.
Differentiation	Correctly computes $dr/d\theta$ using appropriate rules (chain, product).	Computes $dr/d\theta$ with minor, non-systematic errors.	Incorrect derivative, impacting the whole integral.	Derivative is incorrect.
Simplification	Simplifies the integrand $r^2 + (r')^2$ flawlessly using trig identities ($\sin^2\theta + \cos^2\theta = 1$).	Simplifies $r^2 + (r')^2$ correctly but misses a trig identity later.	Minor errors in algebra or trig simplification.	Cannot simplify the expression inside the root.
Evaluation	Correctly evaluates the definite integral, finding the final exact length.	Sets up integral correctly but makes an error in integration technique.	Attempted integration, but fails to complete evaluation.	Cannot evaluate the final integral.